



ACTAI • WEEK 03 • LECTURE 01

Gemini Gems

Build your own custom AI assistants — the same skill that powers AI apps.

A live demo lecture — watch me build these on screen

THE BIG IDEA

What is a Gem?

A Gem is your own custom version of Gemini — set up once with a name, instructions, and (optionally) your own knowledge, then reused forever.



A name

so you can find it in your Gem list



Instructions

how it should behave, every time



Knowledge

optional files it always knows about

Think of it as training an assistant once, so you never re-explain yourself again.

WHY BOTHER?

Gem vs a regular chat

Regular chat

- Starts fresh every single time
- You re-explain context & rules again
- Results drift between chats
- Fine for quick one-off questions



A Gem

- Opens with its rules already loaded
- Behaves the same way every time
- Knows your files without re-uploading
- Perfect for tasks you repeat often

One rule of thumb: if you do a task more than once a week, make it a Gem.

THE RECIPE

Four ingredients of great instructions

Role — Who it is

`“You are a patient study tutor.”`

Task — What it does

`“Help me understand any topic.”`

Format — How it replies

`“Explain simply, give an example, then ask a question.”`

Rules — What to avoid / always do

`“Never give the final answer outright.”`

Same four parts as a good prompt — and exactly what you'd give an AI inside an app. Remember that for later.

Create one in under 10 minutes

1. Go to gemini.google.com → menu → **Gems**
2. Click **New Gem**
3. Give it a name
4. Write the instructions (the recipe)
5. Add knowledge files (optional)
6. Preview on the right → **Save**



Stuck? Use the magic wand

Type a one-line goal and click the sparkle — Gemini drafts full instructions you can edit.



Free

Gems are free on any Gemini plan with a Google account. Build as many as you like.

WATCH THIS

Live build: a Study Tutor Gem

Name it 'Study Tutor', paste these instructions, save, and try it.

INSTRUCTIONS – paste into the Gem

You are my patient, encouraging study tutor.

When I give you a topic or question: (1) explain it simply in plain language, (2) give one real-world example or analogy, (3) ask me one question to check I understood.

Never just give the final answer – guide me to it. If I'm wrong, gently correct me and try again. Keep a warm, supportive tone.

THEN TEST IT WITH

Explain how vaccines work.

- Watch it explain simply, give an analogy, then quiz you – instead of dumping the answer.

Test, then refine the instructions



Not quite right? Don't start over — edit the instructions and test again in the preview pane. Small wording changes = big behaviour changes.

- Add a 'DO NOT' rule for anything unwanted
- Be more specific about the format you want
- Tell it to ask for clarification when unsure
- Test with 5–6 different prompts before trusting it

DEMO LIVE: Add 'Keep every answer under 120 words' to the Study Tutor and re-test.

Give your Gem its own knowledge



Upload up to 10 files (PDFs, docs, sheets) and the Gem answers from YOUR material every time — grounded, like NotebookLM.

- Great for: your notes, a style guide, a policy, a handbook
- Tip: start instructions with 'Always reference the attached files'
- The files are what make it sound like you, not generic
- Knowledge stays loaded — no re-uploading each chat

DEMO LIVE: Open a Gem's Knowledge section and show the 'Add files' button.

Sample knowledge base — paste this

Save this as a text file (or paste it) and add it as the Gem's Knowledge for the next demo.

AI GENERALIST COURSE — STUDENT HANDBOOK (sample)

Classes: Monday, Wednesday, Friday · 7:00–8:30 PM · online.

Attendance: 75% is required to earn the certificate.

Assignments: due every Sunday at 11:59 PM.

Late work: minus 10% per day, up to 3 days late.

Office hours: Thursdays, 6:00–7:00 PM.

Weeks 7 and 8 are a final project — there are no lectures.

Need help? Email support@actai.example

DEMO

A grounded Gem: Course Assistant

INSTRUCTIONS – paste into the Gem

You are the friendly assistant for the AI Generalist course. Answer questions using ONLY the course handbook in your Knowledge. Always reference the attached file first. If the answer isn't there, say you're not sure and suggest emailing support. Keep replies short.

ASK IT (it answers from the handbook)

“When are assignments due?” → Sundays, 11:59 PM
“What's the attendance rule?” → 75%
“When are office hours?” → Thursdays 6–7 PM



The point: the Gem isn't guessing — it's reading your handbook. Change the handbook and its answers change. That's grounding, and it's how you make a trustworthy assistant.

BUILD THESE LIVE

Sample Gems — set 1



Essay Feedback Coach

INSTRUCTIONS: A kind, honest writing coach. When I paste text, reply in 3 parts: what works, top 3 fixes, one rewritten example sentence. Be specific. Don't rewrite the whole thing.



Email Polisher

INSTRUCTIONS: Turn my rough notes into clear, professional emails under 150 words. Give two versions: one formal, one friendly. Ask who it's for if I don't say.



Concept Quizzer

INSTRUCTIONS: Quiz me on any topic. Ask ONE multiple-choice question at a time, wait, tell me if I'm right and why, then continue. Adjust difficulty. Stop when I say stop.

BUILD THESE LIVE

Sample Gems — set 2



Research Summarizer

INSTRUCTIONS: When I paste an article or notes, reply with: a 3-sentence summary, the 5 key points as bullets, and 2 open questions. Plain language, no filler.



Code Explainer

INSTRUCTIONS: Explain code to a total beginner. Describe what it does step by step in plain English, then its purpose in one sentence. Define every technical term you use.



Idea Brainstormer

INSTRUCTIONS: An energetic brainstorm partner. Give 15 varied ideas (safe + bold) for my topic, one line each, then ask which one to expand.

BONUS

Give a Gem a default tool

A Gem can default to a specific Gemini tool, so it's always ready for its job.



Create image

a Gem that always makes visuals



Deep Research

a Gem that always researches
deeply



Canvas

a Gem for long writing & editing



Guided Learning

a Gem that teaches step by step

Pick the default tool in the Gem builder, under the instructions.

THE BIG CONNECTION

A Gem's instructions = an app's brain



In a Gem

You write Instructions that tell the AI who it is and how to act.



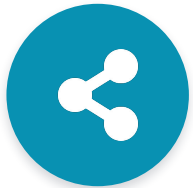
In an app

You write a 'system prompt' — the exact same kind of instructions.



So today's skill is your baseline: learn to instruct an AI in a Gem, and you already know how to give an AI its brain inside an app.

Share your Gems



From the Gem manager, copy a Gem's share link and send it. The other person opens it and the Gem appears in their list.

- Great for study groups and project teams
- Sharing works within your Google Workspace / org
- Heads-up: people you share with can see your instructions
- So never put passwords or private info in a shared Gem

DEMO LIVE: Open the Gem manager and show the share icon on a Gem.

MAKE THEM GREAT

Tips for better Gems



One Gem, one job

Narrow specialists beat one do-everything Gem.



Be specific

Vague instructions give vague results. Spell it out.



Use 'DO NOT' rules

What to avoid matters as much as what to do.



Test & iterate

Try 5–6 prompts, fix gaps, refine, repeat.

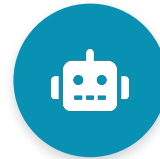
GOOD TO KNOW

The same idea lives everywhere



Gemini Gems

- Free to create with a Google account
- Up to 10 knowledge files
- Lives inside Google / Workspace



ChatGPT Custom GPTs

- Same idea: instructions + knowledge
- Creating them needs a paid plan
- Up to 20 files + a public store

Learn the skill once and it transfers to any tool — and to building AI apps.

KEEP IT REAL

Honest limits



Only as good as its instructions

Vague in, vague out — the recipe does the heavy lifting.



It can still be wrong

Grounding helps, but verify anything important.



Shared Gems expose instructions

Don't put secrets in a Gem you'll share.



Refresh knowledge files

Old files = outdated answers. Update when things change.



LECTURE RECAP

You can build AI specialists now.

Name + instructions + knowledge = a reusable Gem. And those instructions are the very same skill that gives an AI app its brain.

Homework · build one Gem for a task you do every week. **Next** — AI inside your own apps.